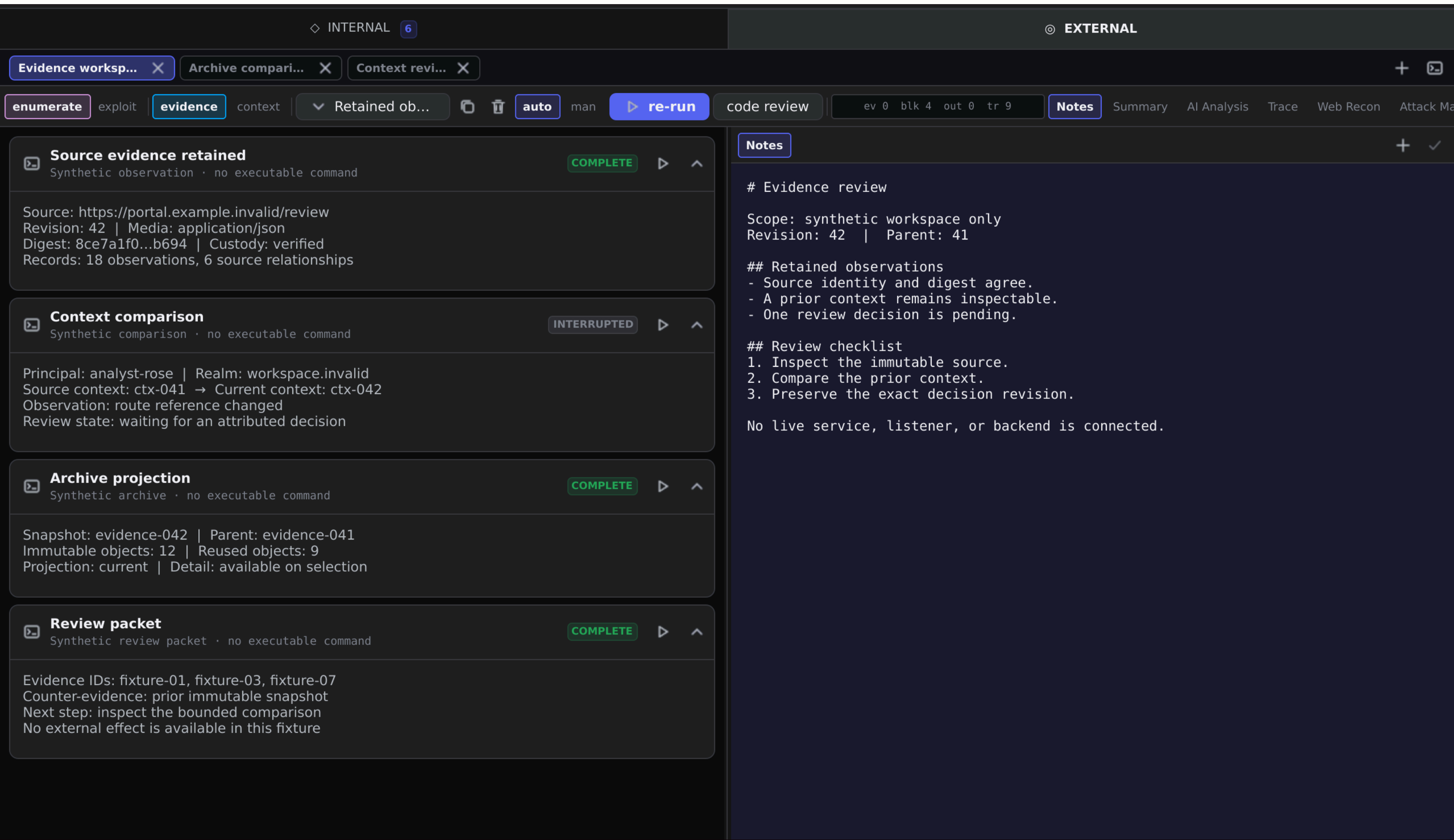
RedShades

Durable security orchestration and native operator control.



Native evidence workspace · Qt capture · synthetic fixture

RedShades is an 88-stage security-assessment system designed to make bounded, short-horizon model work accumulate into a coherent long-horizon result. Deterministic scheduling retains state, ownership, evidence and recovery; supervised AI is invoked only where the next decision cannot be reduced safely to fixed logic.

Prepared for technical leadership at Cybears, Algeria.

The 88-stage deterministic spine

Enumeration, attack-surface construction, controlled validation, foothold handling, pivoting, platform-specific post-exploitation, privilege escalation and reporting are joined by one evidence-driven scheduler rather than one fragile model transcript.

`ENUM` **Discovery and enumeration**

Network, service, web and identity observations

`FOOT` **Foothold convergence**

Evidence correlation, candidate review and validated effect

`PIV` **Pivot and subchains**

Route-aware discovery and child execution graphs

`PRIV` **Privilege progression**

Platform-specific evidence and bounded actions

`C2` **Implant-driven continuation**

Typed sessions, callbacks and host capabilities

`RPT` **Evidence and reporting**

Retained findings, source coordinates and outcomes

Eighty-eight stages share one durable frontier; only eligible work advances.

A01 Authoritative Engagement Architecture · A07 Canonical Specialist Contracts · A11 Objective Review Contracts · C02 Archive Capture Integrity · Q04 Investigation Graph and Workbench · G04 Reporting Workflow · G05 RedShades Canonical Vocabulary

Executive supervision over deterministic work

The scheduler owns ordinary progression. The Executive observes the frontier, gates and evidence stream, resolves cross-stage ambiguity, preserves priorities and routes only the exceptional last mile to ACP or AEP. It does not replace deterministic authority.

WFS

Work Frontier Scheduler

Selects evidence-eligible stages

EXEC

Executive

Observes convergence and routes exceptional work

ADS

Bounded AI calls

Fill local ambiguity inside typed stages

ACP

Adaptive Capability Pipeline

Synthesizes a missing target interaction

AEP

Autonomic Engineering Plane

Repairs defective production machinery

TAC

Transactional Action Coordinator

Retains intent, effect and read-back evidence

The Executive monitors the chain; deterministic code retains authority.

A01 Authoritative Engagement Architecture · A07 Canonical Specialist Contracts · A08 Specialist Adapters · A13 Host Execution Attribution · R05 Isolated Stage Workers · G05 RedShades Canonical Vocabulary

Model allocation: spend intelligence where continuity matters

The architecture separates small bounded decisions from persistent convergence. Short calls can be served by a shared compact model with boundary-specific QLoRA specialization; Executive, ACP and AEP work benefits from persistent sessions on the stronger fine-tuned model. This is the intended allocation architecture, not a claim that every planned adapter has completed training.

1B

Shared compact base

Classification, normalization and narrow stage decisions

QLR

Boundary QLoRA bank

Planned specialist adaptation without duplicating the base

30B2A

Fine-tuned reasoning model

Persistent Executive, ACP and AEP sessions

KV

Session continuity

Prior rejection receipts and causal history remain available

DET

Deterministic custody

Scope, tools, effects, validation and promotion remain outside the model

REV

Isolated review

Independent contamination and semantic assessment

Small models handle bounded work; persistent stronger sessions own convergence.

G01 Component Architecture · R01 Runtime Composition Kernel · R02 Runtime Lifecycle Adoption · R03 ACP Vertical Session · R04 Autonomic Engineering Plane Contract · D01 Deployment and Process Ownership · E04 CI Evidence and Cache Custody · A12 Specialist Plugin Registry · G05 RedShades Canonical Vocabulary

Operational surface and native control

The chain coordinates established assessment tools, custom workers, Go implants, Zig stage helpers, payload bundles, virtualized transformation layers and native Qt inspection surfaces. Tools remain replaceable executors behind typed observations rather than becoming the architecture itself.

NET

Network and service

Nmap, Masscan, protocol clients and route-aware discovery

WEB

Web assessment

FFUF, Gobuster, Nuclei, Gowitness, SQLMap and browser evidence

ID

Identity and directory

NetExec, Impacket, BloodHound, LDAP and SMB tooling

PIV

Pivoting

Ligolo, Chisel and nested route-aware scans

IMP

Implants and helpers

Go Windows/Linux implants and Zig stage-zero helpers

UI

Native operator workspace

AttackMap, Network Graph, WebRecon, evidence and gates

A broad executor ecosystem is coordinated through typed state and operator-visible evidence.

CES

Canonical Engagement State

Canonical Engagement State (CES) links evidence, identity, authority, actions, resources, and recovery records in one revision. Validation requires reference closure and permitted transitions before persistence. Credentials retain provenance and applicability, while related artifacts form digest-bound sets.

CES

Canonical Engagement State

Evidence + identity
typed inputs

→

Complete revision
validated references

→

Derived projections
Qt · graph · reports

INPUT
Linked engagement records

OUTPUT
One validated revision

AUTHORITY BOUNDARY
Canonical domain owns state

Implemented contract · A01

A decision uses one coherent revision.
CONCEPTUAL DIAGRAM

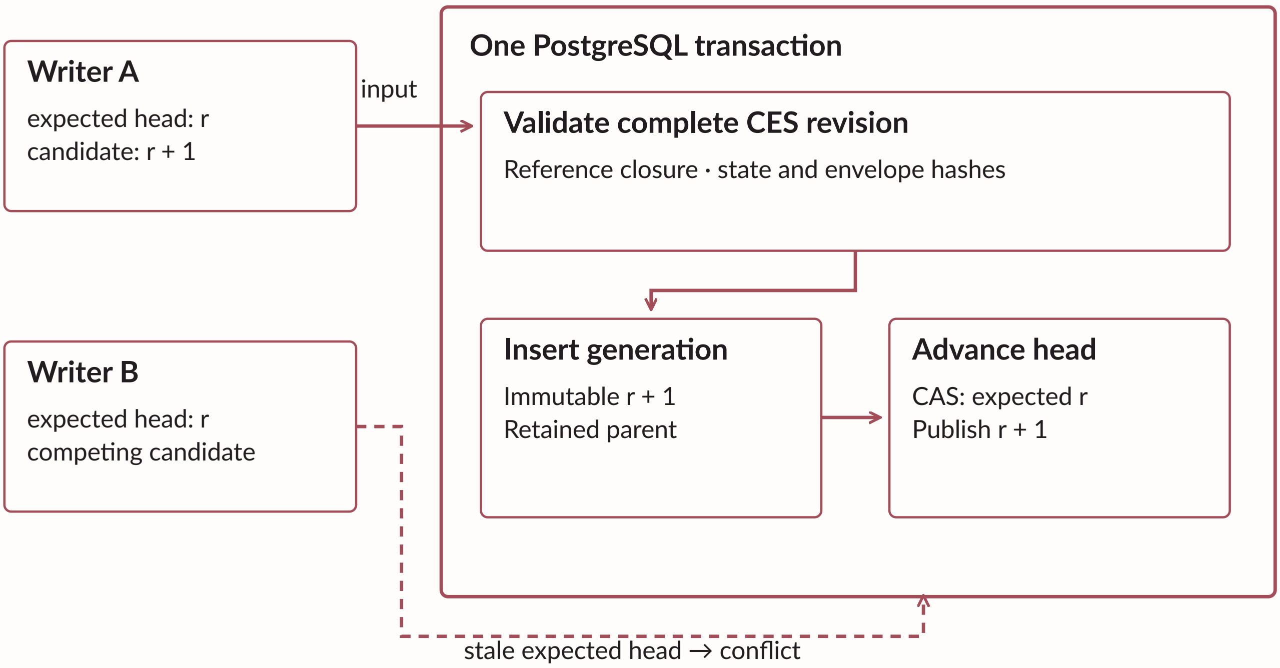
IGS

Immutable Generation Store

The Immutable Generation Store (IGS) validates the proposed state, checks the expected head, and commits a complete immutable generation with its new head atomically. Separate state and envelope hashes bind content and lineage. Repeated idempotency keys require identical payloads.

CES and IGS: one complete revision

Canonical Engagement State and Immutable Generation Store share an atomic boundary.



CAS: compare-and-swap · a stale writer cannot silently replace the current head

Competing writers receive an explicit conflict.

CONCEPTUAL DIAGRAM

Lifecycle and interruption

Typed events drive lifecycle reduction. Completed, failed, and cancelled outcomes remain terminal. A nonterminal process with no runtime owner becomes interrupted; an effect that may already have occurred retains an unknown outcome until observation or reconciliation resolves it.

CONDITION	RECORDED MEANING
Queued	Work exists; execution has not begun.
Running	A current owner holds the attempt and progress lease.
Interrupted	Nonterminal work has lost its runtime owner.
Unknown outcome	An effect may have occurred; observation or reconciliation is required.
Terminal	Completed, failed, or cancelled retains its final outcome.

Interruption and unknown outcome preserve different recovery obligations.
CONCEPTUAL DIAGRAM

A05 Execution Graph Lifecycle · A07 Canonical Specialist Contracts

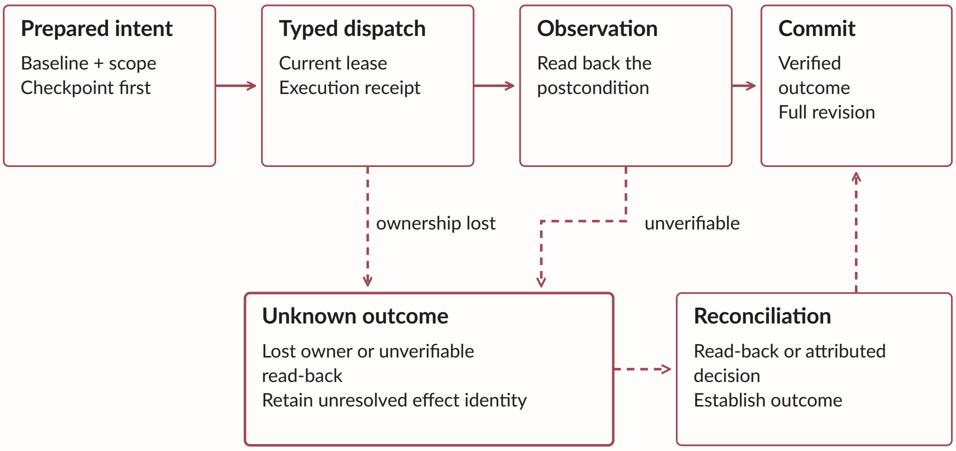
TAC

Transactional Action Coordinator

The Transactional Action Coordinator (TAC) checkpoints intent and baseline before dispatch. A typed executor performs the bounded operation, then an independent read-back checks the postcondition. Lost ownership leaves the action unresolved rather than authorizing a repeated effect.

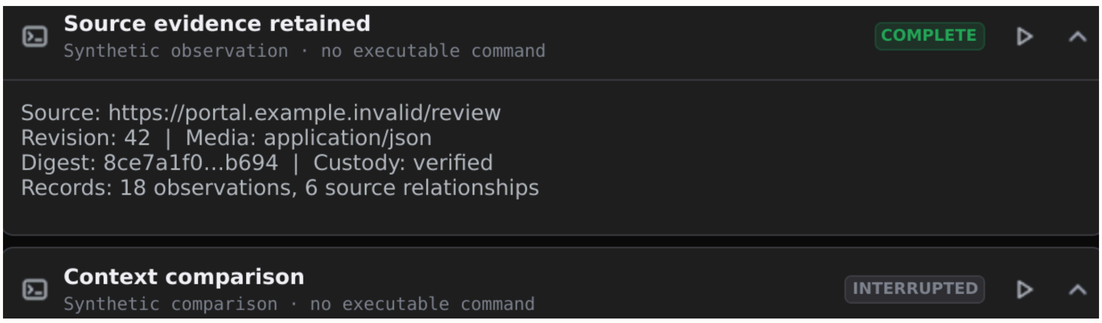
TAC: retain intent and outcome custody

Dispatch success and observed postcondition are different evidence records.



TAC · Transactional Action Coordinator: uncertainty does not authorize a blind retry.

Solid arrow: declared transfer or transition · dashed arrow: conditional branch



Retained observation history · Qt capture · synthetic fixture

Execution receipts and observed outcomes retain separate custody.

CONCEPTUAL DIAGRAM

EVC

Evidence Verification and Custody

Evidence Verification and Custody (EVC) stores raw content by digest and records its size, media type, and provenance. Compact projections carry references. Before source navigation, the backend verifies the retained object and its bounded coordinates. Shared objects require reference-aware reclamation.

EVC

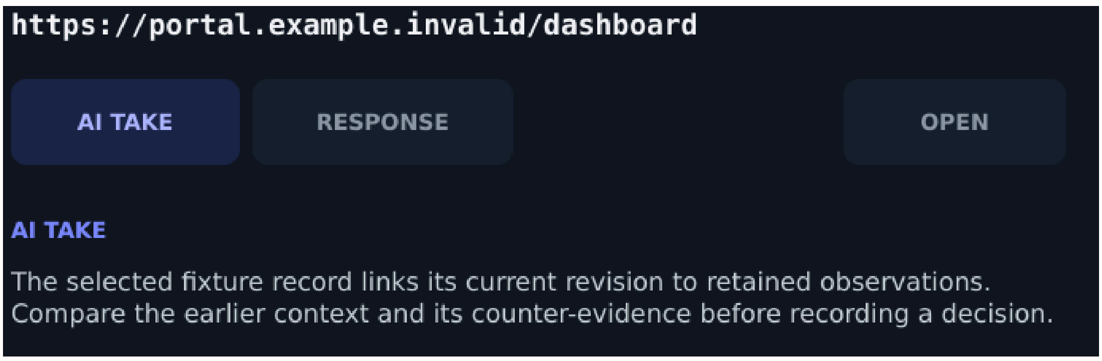
Evidence Verification and Custody

INPUT
Raw artifact and provenance

OUTPUT
Digest and bounded selection

AUTHORITY BOUNDARY
Project-scoped custody

Implemented contract · A07 / C02



Selected evidence detail · Qt capture · synthetic fixture

Selected evidence remains linked to verified source bytes.

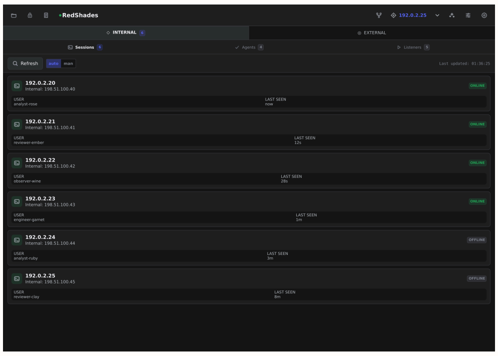
CONCEPTUAL DIAGRAM

A07 Canonical Specialist Contracts · C02 Archive Capture Integrity

ECT

Execution Context Timeline

The Execution Context Timeline (ECT) orders factual frames and causal transitions. Identity, host, process, isolation boundary, session, route, scope, and time retain observed, unavailable, stale, or conflicted status. A changed context returns continuation to review.



OVERVIEW

Synthetic sessions retain host identity, user and current status.

DETAIL

Qt capture · synthetic fixture

FACTUAL FRAME	RETAINED DIMENSIONS
Identity	Principal, host and session
Context	Route, scope and validity
Transition	Observed change and cause

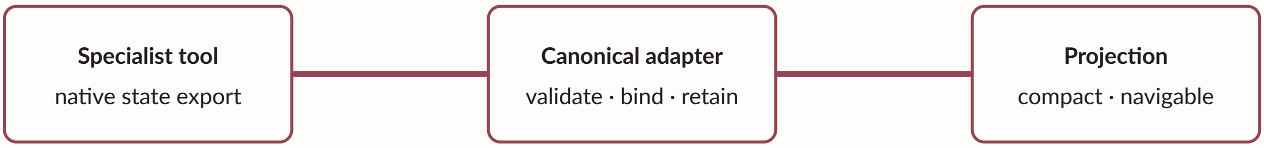
Missing context stays explicit.

QT CAPTURE + CONTEXT MATRIX

SIC

Specialist Interchange Contracts

Specialist Interchange Contracts (SIC) bind sessions, results, selections, annotations, recipes, and checkpoints to exact revisions and source evidence. Five broad adapter families preserve their native fidelity. Unsupported fields or unresolved coordinates remain visible compatibility information.



Different tools share a contract without losing source detail.

CONCEPTUAL DIAGRAM

INPUT

Specialist output and versioned selections

OUTPUT

Typed result and compact projection

AUTHORITY BOUNDARY

Version and custody validation

IMPLEMENTED CONTRACT

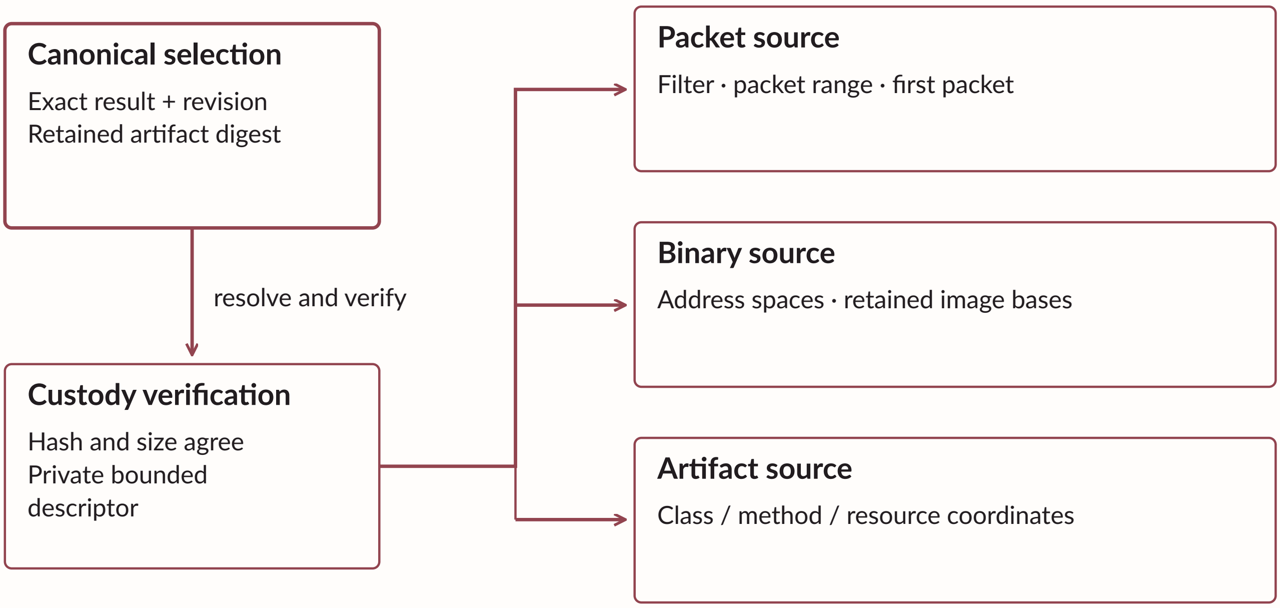
A07 Canonical Specialist Contracts · A08 Specialist Adapters

Source coordinates and viewer handoff

A source handoff starts with a canonical result identity. Packet selections retain filters and ranges; binary records preserve address spaces and image bases; artifact records retain source coordinates. Private same-user descriptors bind the verified artifact to bounded viewer input.

Source fidelity survives the viewer handoff

EVC verifies retained bytes; SIC preserves inspection coordinates.



Viewer handoff preserves source-tool provenance. Deep-link support remains explicit.

Solid arrow: declared transfer or transition · dashed arrow: conditional branch

Viewer support determines which retained coordinates can be opened directly.

CONCEPTUAL DIAGRAM

DTE

Deterministic Transformation Engine

The Deterministic Transformation Engine (DTE) preserves ordered plans, implementation versions, input/result hashes, and baseline deltas. Derived observations retain rejected or unmatched inputs and source relationships. The backend accepts a bounded declarative operation catalog.

01

Bound inputs

Exact source records and their revisions

02

Versioned transform

Deterministic operation and parameters

03

Derived result

Typed output with bounded custody

04

Derivation history

Inputs, results, deltas, and retained lineage

A conclusion remains traceable through its derivation.

CONCEPTUAL DIAGRAM

INPUT	OUTPUT	AUTHORITY BOUNDARY
Exact input references and transform parameters	Derived records and retained lineage	Deterministic operation; source remains unchanged

IMPLEMENTED CONTRACT

A07 Canonical Specialist Contracts · Q04 Investigation Graph and Workbench · E03 691 Verification Record

Checkpointed semantic experiments

A semantic experiment binds a recipe, source revision, context, adapter version, scope, and mutation preview. Every completed step writes a checkpoint before its successor begins. Missing inputs wait; changed inputs conflict before dispatch. Protected values remain ephemeral.

01

Bind the plan

Recipe, source, context, scope, and preview

02

Resolve prerequisites

Inputs, adapter availability, and exact approval

03

Checkpoint each step

Retain every verified completed result

04

Recover the prefix

Preserve completed work; reconcile interruption

Recovery retains the verified prefix of an interrupted experiment.

CONCEPTUAL DIAGRAM

A07 Canonical Specialist Contracts · A08 Specialist Adapters

WFS

Scheduling eighty-eight stages without losing the frontier

Each stage declares prerequisites, evidence inputs, effects, outputs, recovery policy and verified postconditions. The scheduler advances independent branches concurrently, parks only blocked endpoints and reconstructs progress from durable state after interruption.

ELIGIBILITY INPUT	REQUIRED CONDITION
Capability	Verified, applicable evidence
Resource	Current health and valid dependency
Context	Exact identity, placement, route, and scope
Decision	Deterministically validated receipt
Lease	Owner and generation; heartbeat, progress, and overall deadlines

A live worker can still be stalled.

CONCEPTUAL DIAGRAM

INPUT

Capabilities, context, resources, decision receipts

OUTPUT

Eligible work and stable blocker records

AUTHORITY BOUNDARY

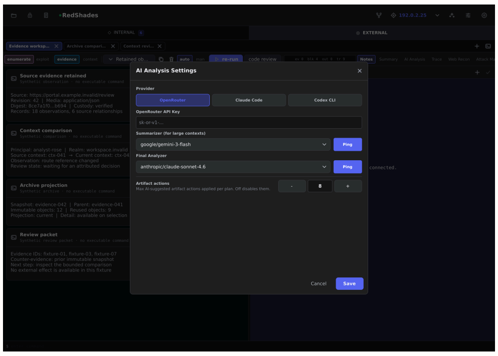
Expected engagement revision

IMPLEMENTED; LEGACY ADOPTION INCOMPLETE

ADS

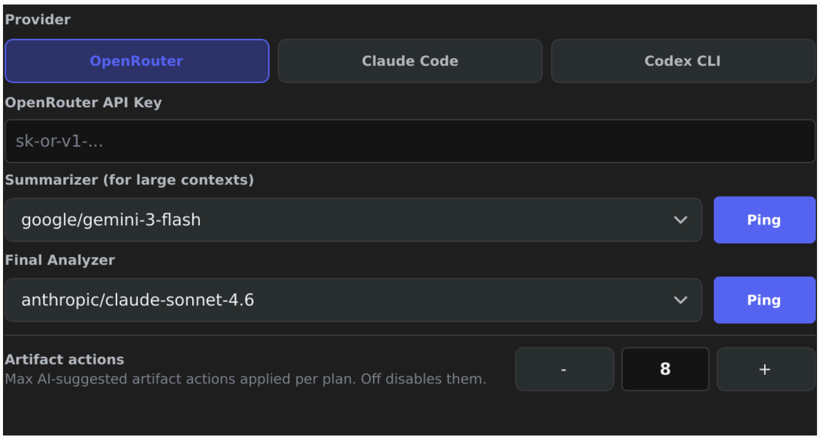
Miniature AI decisions inside deterministic stages

A stage invokes AI only for bounded ambiguity: ranking evidence, normalizing an unfamiliar label, selecting admitted mechanics or reviewing a compact candidate. The response becomes runtime truth only after deterministic validation.



OVERVIEW

Provider selection, bounded model configuration and artifact-action limits.



DETAIL

Qt capture · synthetic fixture

ADS

AI Decision Service

Evidence packet — revision-bound → **Provider result** — bounded work → **Validation** — typed contract → **Decision receipt** — durable record

INPUT

Evidence and typed request

OUTPUT

Validated decision receipt

AUTHORITY BOUNDARY

Provider output has no authority

Implemented; legacy adoption incomplete · A01

Validated decisions return through application authority.

CONCEPTUAL DIAGRAM

A01 Authoritative Engagement Architecture

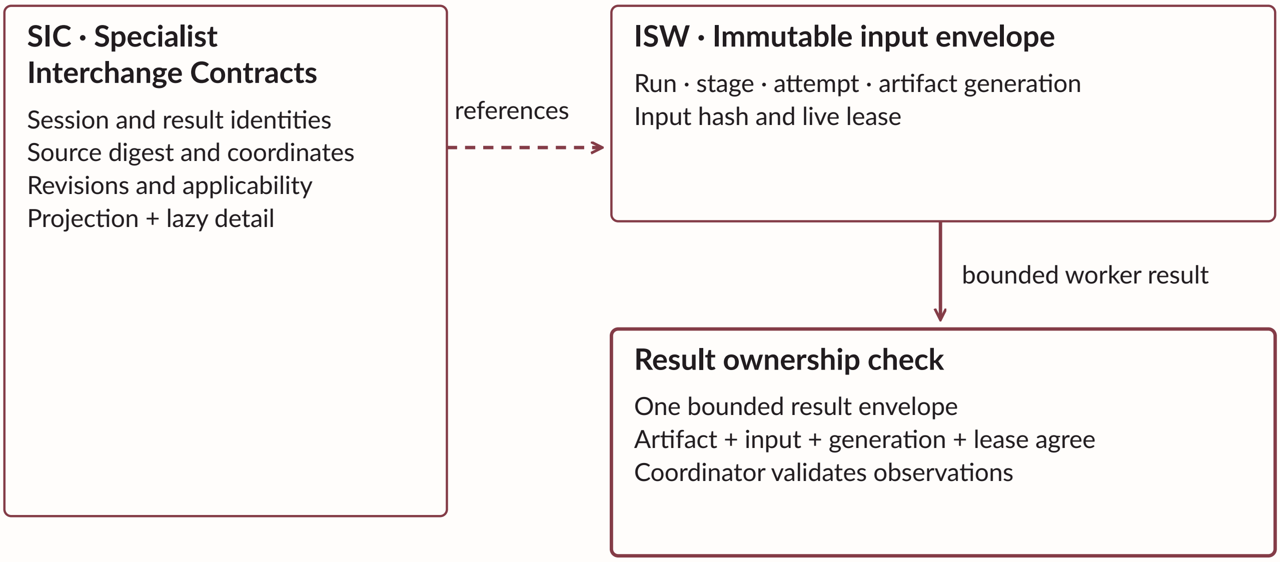
ISW

Isolated Stage Workers

Isolated Stage Workers (ISW) receive immutable envelopes and return bounded result envelopes. A result can commit only under the current artifact, input hash, generation, and live lease. Declared outputs require coordinator-owned validation. Legacy in-process handlers wait for a scheduler boundary.

SIC and ISW: records and worker ownership

Specialist references and isolated execution retain separate contracts.



SIC preserves fidelity. ISW owns isolated execution boundaries.
Stale results are fenced; legacy handlers wait for a scheduler boundary.

Dashed arrow: declared input references when applicable · ISW: Isolated Stage Workers

Replacement fences stale results before they can commit.

CONCEPTUAL DIAGRAM

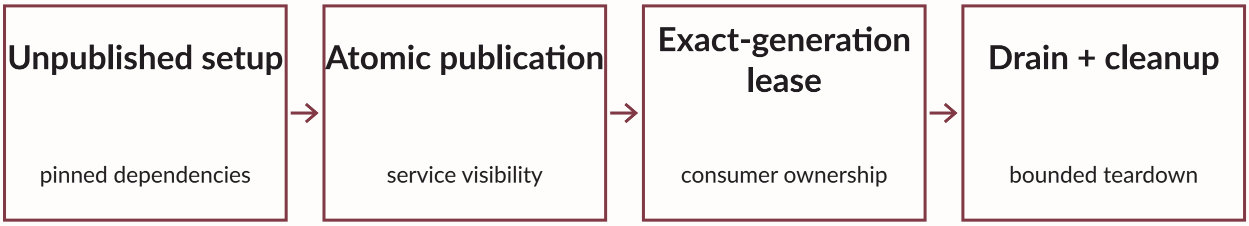
RCK

Runtime Composition Kernel

The Runtime Composition Kernel (RCK) prepares providers in unpublished scopes, pins dependencies, and records effects before publication. Consumers lease exact generations. Failed setup unwinds its effects, and cleanup executes once even when several callers wait.

RCK

Runtime Composition Kernel



INPUT

Typed provider and manifest

OUTPUT

Published generation and leases

AUTHORITY BOUNDARY

Canonical state stays external

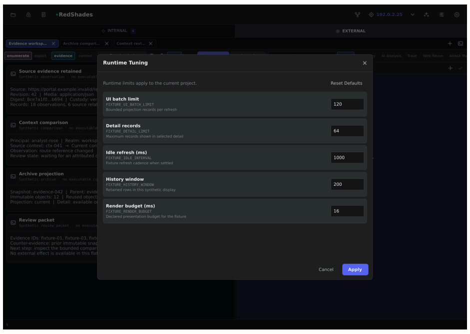
Implemented contract · R01

Provider ownership stays separate from engagement state.

CONCEPTUAL DIAGRAM

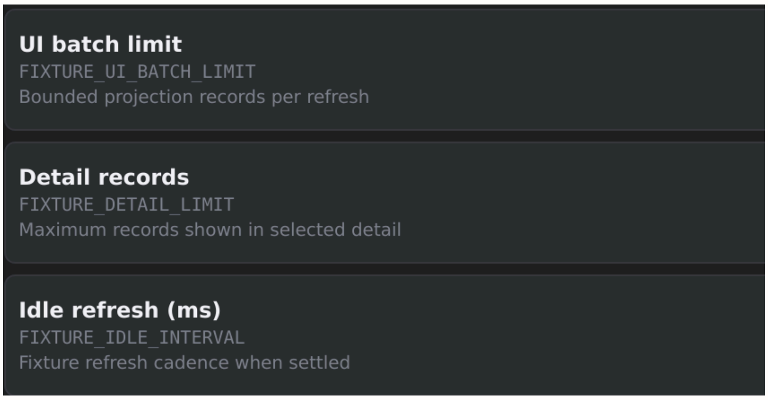
Provider generations and switchboard

The switchboard retains provider identity, fencing, routing, and process custody. New calls resolve the published generation; existing calls stay pinned until release. Candidate state transfer is validated before publication. Rollback changes provider availability while preserving external outcome history.



OVERVIEW

Native controls expose bounded runtime and rendering settings.

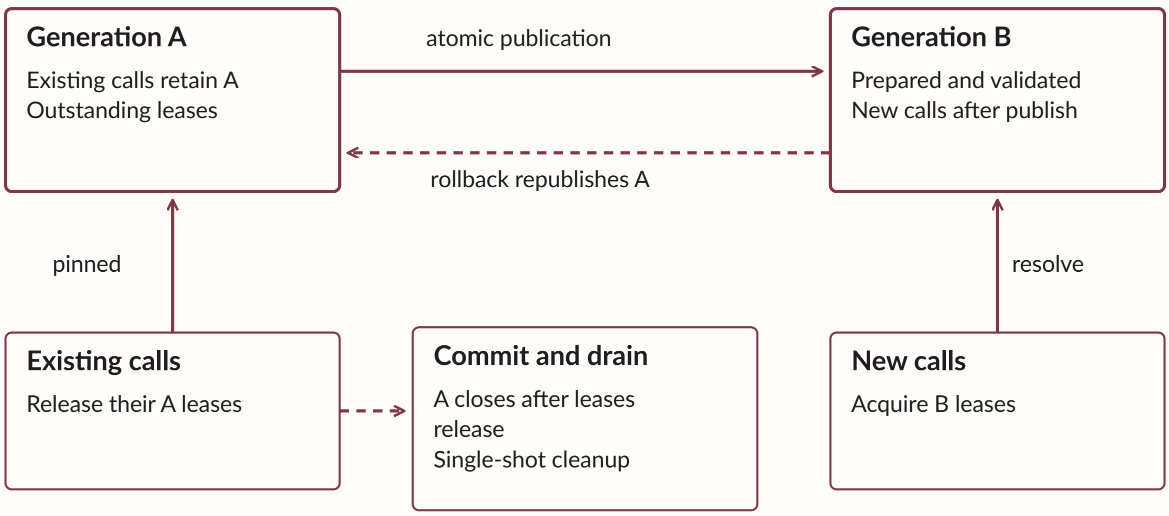


DETAIL / FIELD LABELS

Qt capture · synthetic fixture

RCK: publish, pin, drain and roll back

Replacement changes availability under exact-generation ownership.



RCK · Runtime Composition Kernel
Provider rollback preserves the history of external effects.

Solid arrow: declared transfer or transition · dashed arrow: conditional branch

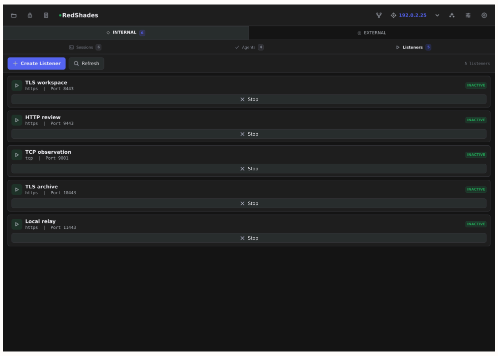
Generation identity survives concurrent replacement.

QT CAPTURE + GENERATION DIAGRAM

R01 Runtime Composition Kernel · R02 Runtime Lifecycle Adoption

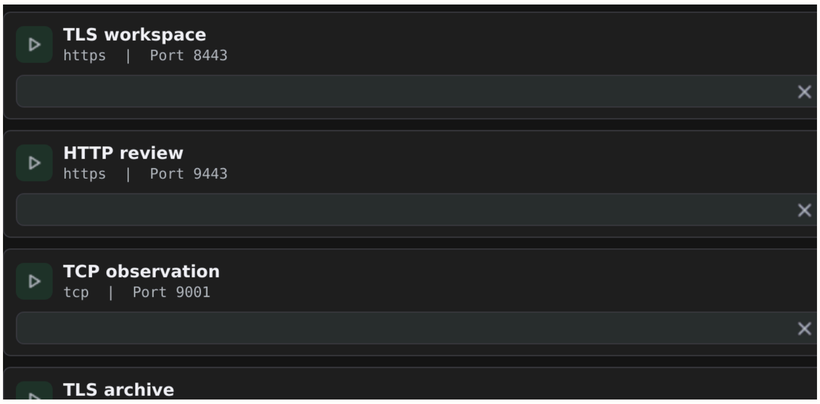
Process ownership and health

Resource supervisors retain desired state, process ownership, current health, backoff, and recovery status. A named bounded health observation admits dependent work. Qt panes may detach from warm specialist windows while their dedicated supervisors retain lifecycle responsibility.



OVERVIEW

Listener projections retain type, binding and visible state.



DETAIL

Qt capture · synthetic fixture

SUPERVISOR FACT	MEANING
Desired / owned	Required resource and current owner
Healthy / applicable	Bounded observation and exact dependency

Process existence alone does not establish health.

QT CAPTURE + OWNERSHIP MATRIX

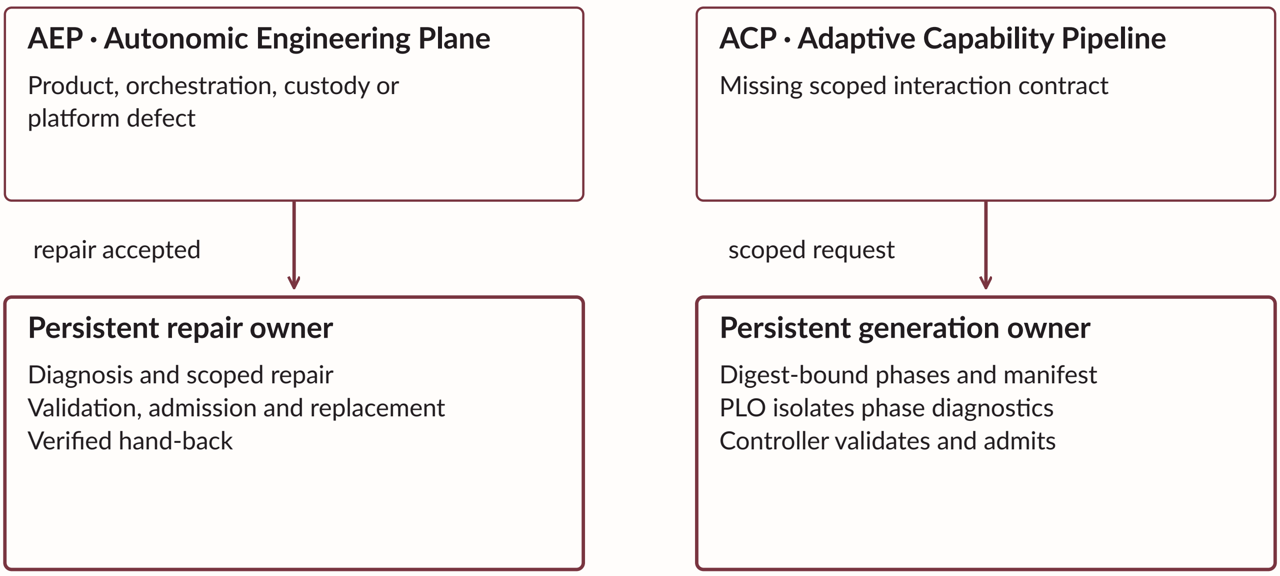
AEP

AEP — repairing the machine without losing the run

When the blocker is a product, orchestration, evidence-custody or platform-capability defect, the Executive routes it to AEP. One persistent repair session owns diagnosis, source work, focused and counterfactual tests, admission, atomic replacement and verified hand-back to the interrupted frontier.

AEP and ACP: separate engineering owners

Routing follows the defect class; neither column is a shortcut to effect authority.



Separate incident classes preserve separate ownership.
Complete AEP convergence remains under validation.

PLO: Phase-Level Optimization · a phase receipt does not establish an external outcome

Complete-loop validation remains in progress.

CONCEPTUAL DIAGRAM

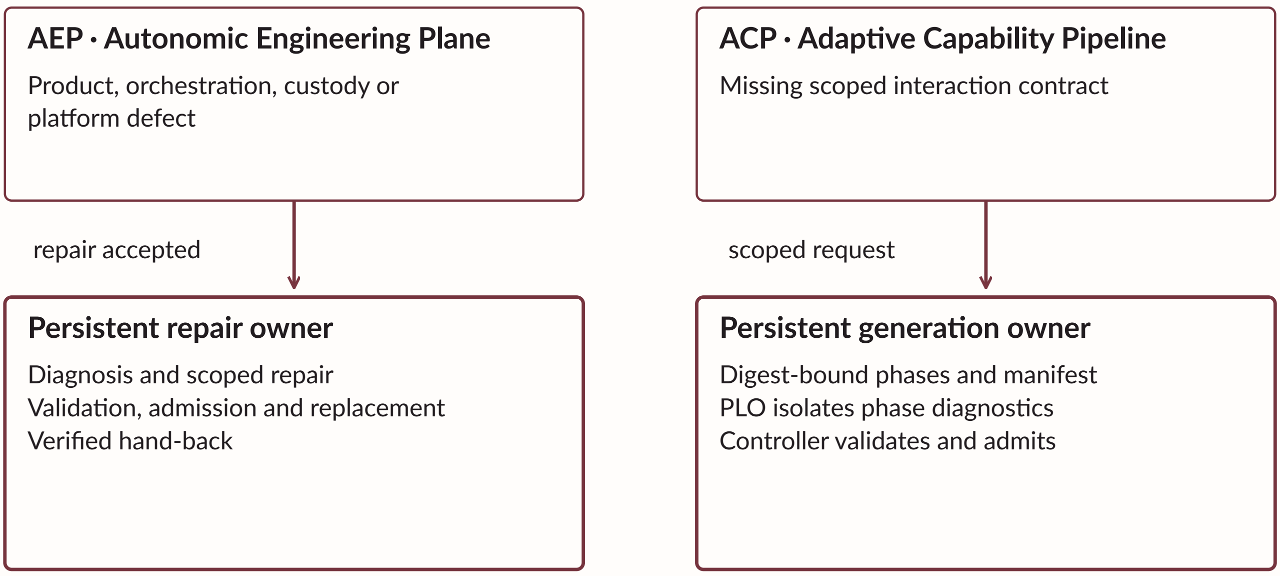
ACP

ACP — solving the unfamiliar last mile

When evidence proves that the chain lacks one target interaction, ACP owns the complete capability convergence loop: finding, contract, binding, compatible SDK selection or extension, adapter generation, tests, isolated review and execution proof. Existing compatible artifacts skip unnecessary boundaries.

AEP and ACP: separate engineering owners

Routing follows the defect class; neither column is a shortcut to effect authority.



Separate incident classes preserve separate ownership.
Complete AEP convergence remains under validation.

PLO: Phase-Level Optimization · a phase receipt does not establish an external outcome

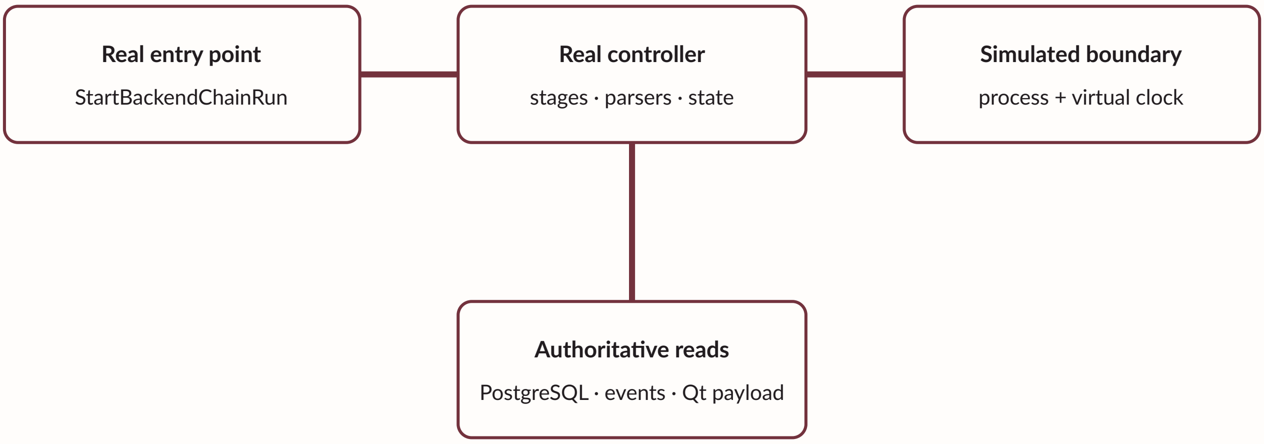
A phase receipt does not establish an external outcome.

CONCEPTUAL DIAGRAM

R03 ACP Vertical Session · G05 RedShades Canonical Vocabulary

One chain: enumerate, converge, pivot and continue

The operational graph can move from external enumeration through web or identity evidence to a validated foothold, register the resulting session, discover internal routes, spawn subchains, evaluate privilege paths and continue through implant-backed host stages. Every transition remains tied to evidence and authority.



Environmental behavior requires separate live validation.

CONCEPTUAL DIAGRAM

V01 Chain Simulator and Validation Ladder · V02 Simulator Architecture · V04 Simulator Live-Escape Audit

CI and cache custody

CI uses isolated checkouts and private scratch with admitted shared build/module caches. Ultra records bounded receipts for eight verification lanes. The latest documented Go run, 1599, failed three root contracts and retained an ambient CLI process tree after passing earlier gates.

Verification receipts bound the claim

CI evidence retains its revision, execution scope and current status.

Verification lane
Exact source revision
Declared checks and bounds
Explicit lane status

→

Retained receipt
Command and status
Bounded output + duration
Artifact hashes

→

Supported claim
This revision and fixture
This lane and environment
Explicit remaining limits

Historical operator measurement
71,462 dirty pixels / 921,600 viewport
Retained fixture and build

Go run 1599: documented status
Three root contracts failed
New pushed result required after repair

A configured lane, historical pass and current passing result are separate evidence states.

Solid arrow: declared transfer or transition · dashed arrow: conditional branch

Repairs await a new pushed result; newer passing operator visual evidence is unconfirmed.

CONCEPTUAL DIAGRAM

E04 CI Evidence and Cache Custody · E05 Operator Frontend Evidence

Verified artifact deployment

Deployment verifies complete artifact sets, content hashes, and paired Go/Qt binary compatibility before publication. Readiness is recorded last. Atomic publication and rollback distinguish complete restoration from an incomplete outcome. Dedicated runtime owners retain live processes.

01

Stage the set

Complete component files in a private incoming set

02

Verify

Manifest, source revision, digest, and readiness

03

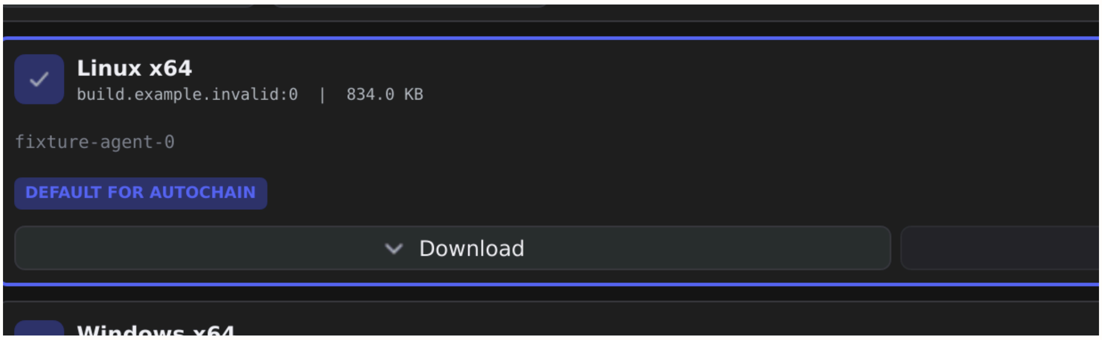
Preflight

Paired Go and Qt ABI compatibility

04

Publish or restore

Atomic replacement with explicit rollback status



Artifact records · Qt capture · synthetic fixture

Artifact admission and process ownership remain independently checked.

CONCEPTUAL DIAGRAM

Rendering and runtime observability

Native graph surfaces preserve geometry and repaint affected regions. The Go bridge polls away from the UI thread and yields between bounded backlog pages. Historical operator verification measured 71,462 dirty pixels within a 921,600-pixel viewport and checked settled-idle behavior.

71,462

dirty pixels

The measurement applies to its retained fixture and build.

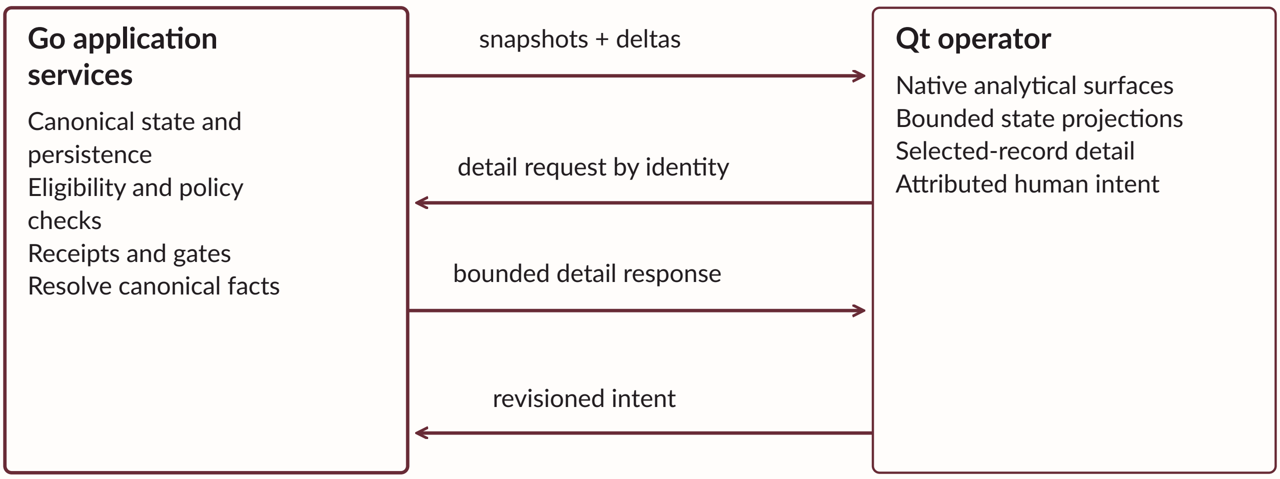
HISTORICAL MEASUREMENT · JULY 2026

Go/Qt interfaces

The Foreign Function Interface (FFI) carries versioned Go/Qt state, lazy detail, and revisioned commands. Application Binary Interface (ABI) checks enforce symbol and signature compatibility. Model Context Protocol (MCP) integrations remain transport boundaries; Go resolves canonical facts.

Go and Qt: state out, revisioned intent in

The native interface separates application authority from operator presentation.



FFI · Foreign Function Interface
ABI · Application Binary Interface: required symbols and signatures checked before pairing.
The UI supplies identity and expected revisions; Go resolves the authoritative record.

Solid arrow: declared transfer or transition · dashed arrow: conditional branch

Views submit intent against an expected revision.
CONCEPTUAL DIAGRAM

G03 Native FFI/API Surfaces · Q01 Qt Operator Architecture

Integrated capability families

The implemented surface combines discovery, web reconnaissance, identity and directory analysis, controlled candidate validation, session registration, pivoting, Windows/Linux host inspection, privilege progression, artifact handling, reporting and autonomous repair.

RedShades responsibility map

Canonical subsystem identities grouped by ownership.

Canonical state and evidence

CES · Canonical Engagement State
EVC · Evidence Verification and Custody
DTE · Deterministic Transformation Engine
IGS · Immutable Generation Store
SIC · Specialist Interchange Contracts
Objective review and derived projections

Context and execution control

ECT · Execution Context Timeline
TAC · Transactional Action Coordinator
WFS · Work Frontier Scheduler
PLS · Progress Lease Supervision
ADS · AI Decision Service
BHG · Bounty Hunting Gate

Runtime ownership and replacement

RCK · Runtime Composition Kernel
ISW · Isolated Stage Workers
Runtime switchboard
Process bundles and resource supervisors

Engineering repair and generation

AEP · Autonomic Engineering Plane
ACP · Adaptive Capability Pipeline
PLO · Phase-Level Optimization
SPR · Specialist Plugin Registry

Operator and diagnostic views

AttackMap · Network Graph · WebRecon
Investigation Graph and Workbench
RCC · Recovery and Continuation Center
TMG · Trigger Methodology Graph
SCG · Shadow Capability Graph
Native Go/Qt bridge

Responsibility bands do not prescribe execution order.

Band outline: responsibility scope · names follow the canonical subsystem glossary

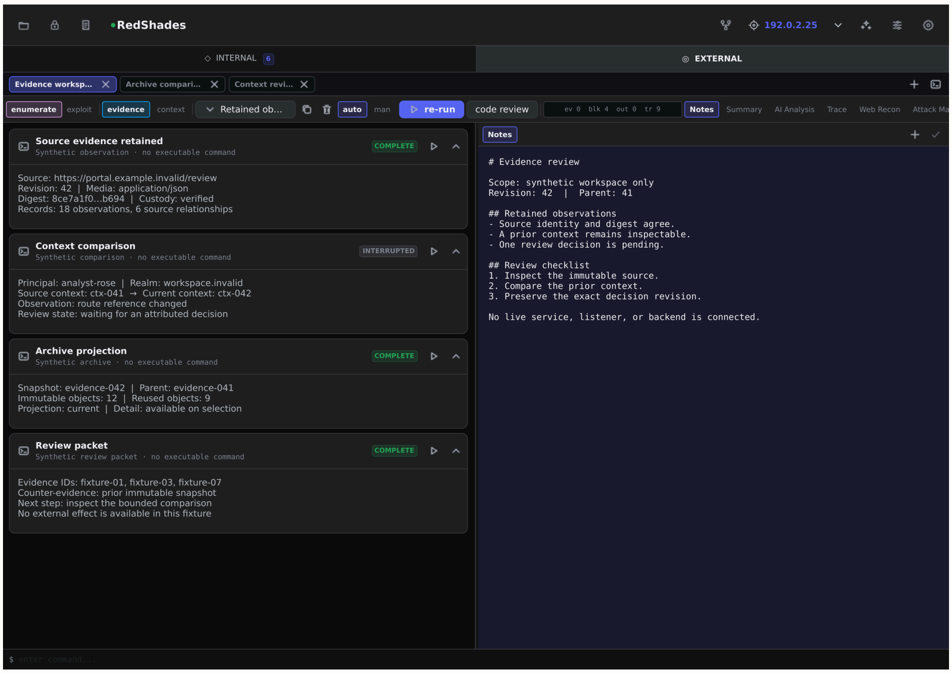
Breadth of interchange exceeds the current breadth of live validation.

CONCEPTUAL DIAGRAM

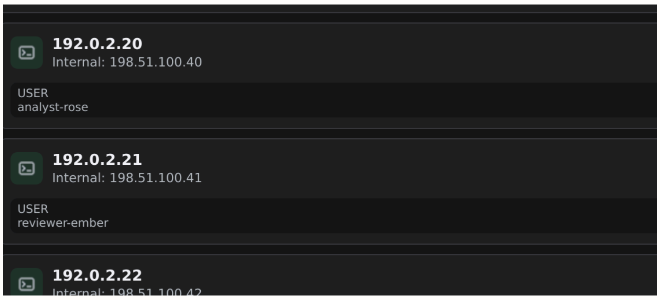
A07 Canonical Specialist Contracts · A08 Specialist Adapters · A12 Specialist Plugin Registry · M01 Trigger Methodology Graph · M03 Shadow Capability Graph · G04 Reporting Workflow

Native operator workspace

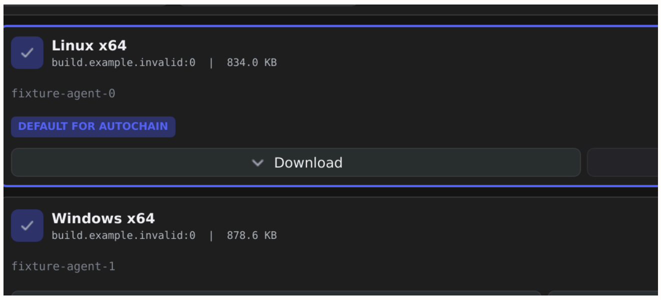
The RedShades operator uses QML for menus, project navigation, and workspace controls. Native QWidget/QPainter surfaces render analytical content. Sessions and external-tool history remain accessible within one window.



Evidence workspace · Qt capture · synthetic fixture



Session identity · detail



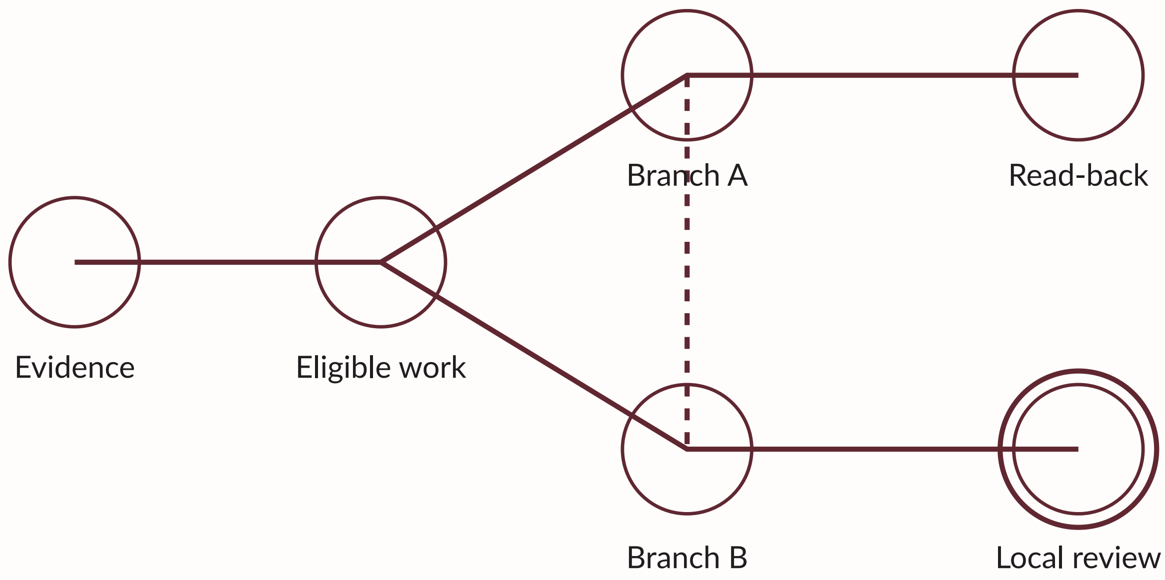
Artifact states · detail

Qt operator workspace with synthetic project and session data.

QT CAPTURES · SYNTHETIC FIXTURES

AttackMap

AttackMap presents plan nodes, runtime instances, dependencies, endpoint lanes, and review state. Go owns their semantics. Stable identities preserve placement as work arrives, and runtime deltas repaint affected regions.



Execution topology connects work to its prerequisites.

CONCEPTUAL DIAGRAM

Canonical node inspection

Node inspection requests exact canonical detail from Go. The response exposes blockers, evidence, current context, tools, attempts, and available actions with their consequences. Detailed streams load separately from the steady-state graph.

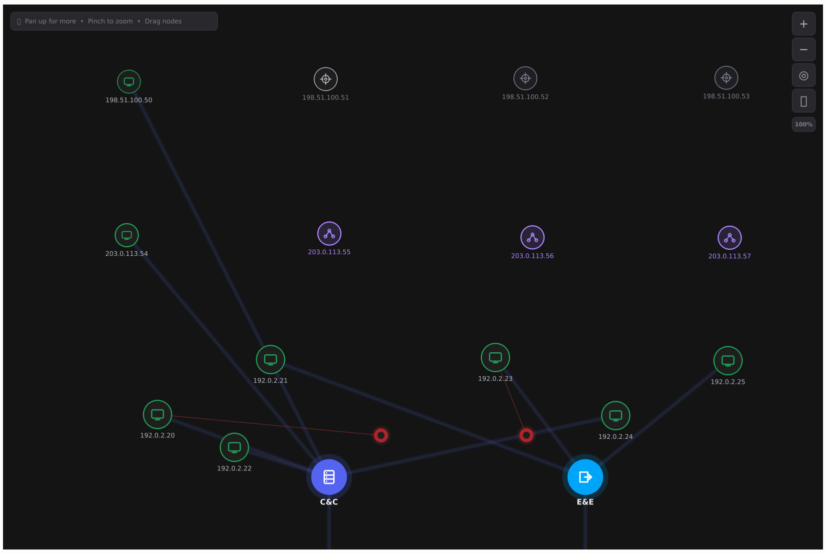
INSPECTION DOMAIN	BOUNDED CANONICAL DETAIL
Identity	Exact plan-node and runtime identities
Evidence	Stable source references and custody
Dependencies	Eligibility and explicit blocker reasons
Context	Principal, host, route, scope and validity
Action	Expected revision, gate and consequences
History	Attempts, checkpoints and cancellation cause

An unavailable action retains a factual reason.

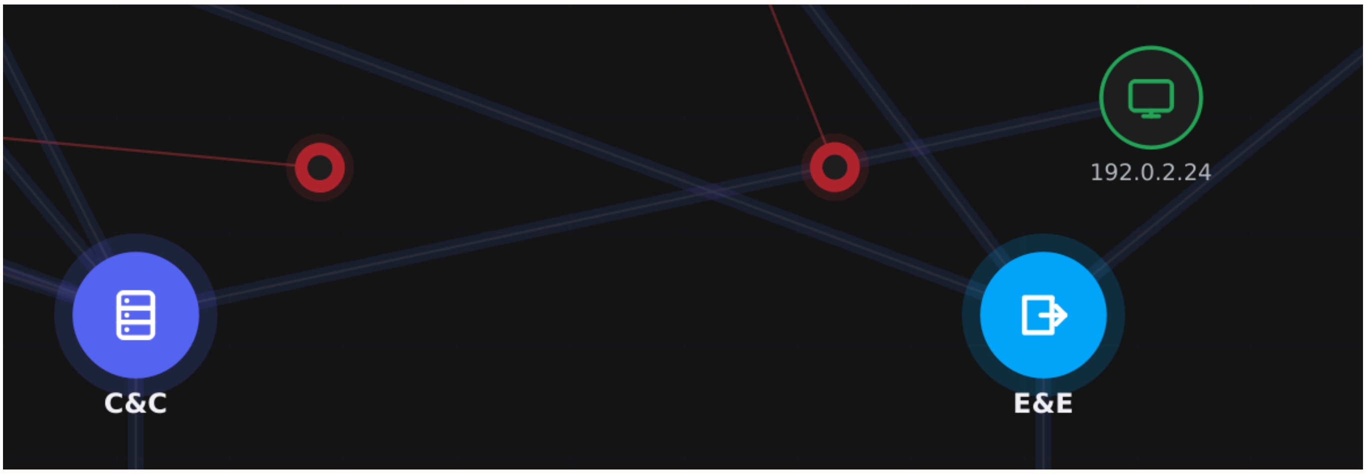
CONCEPTUAL DIAGRAM

Network Graph

Network Graph renders observed hosts, sessions, relays, routes, and relationships. Context navigation matches exact frame references to current node identities. Unavailable topology remains unresolved while factual detail stays accessible in the timeline.



Observed topology · Qt capture · synthetic fixture



Host and session detail · same capture

Network Graph with synthetic host and session data.

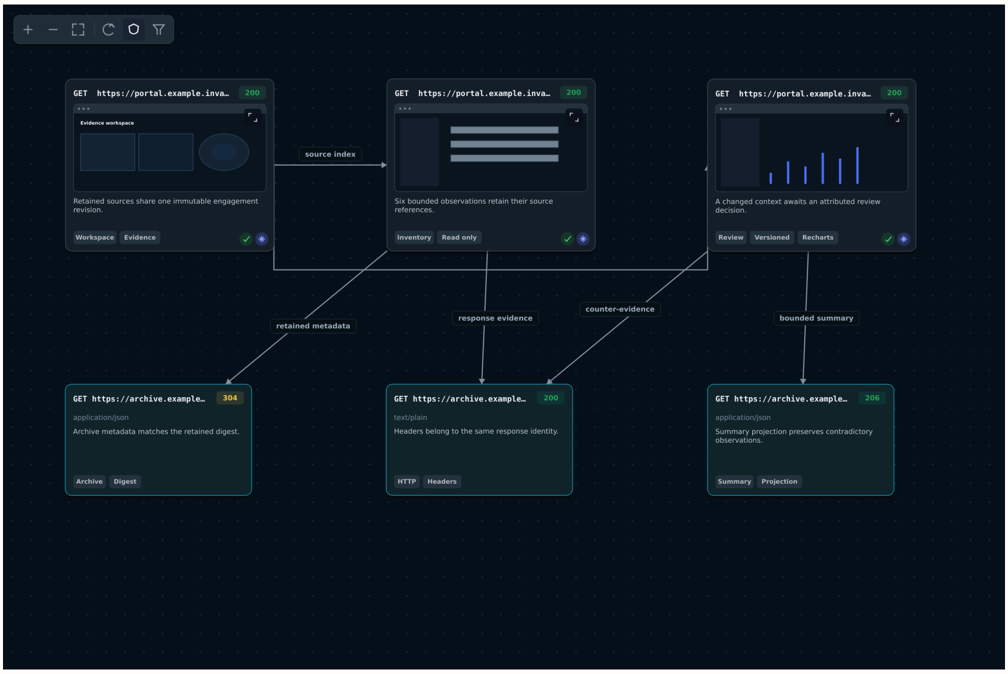
QT CAPTURE · SYNTHETIC FIXTURE

Identity	Context	Absence
Exact graph node	Observed frame reference	Unresolved representation

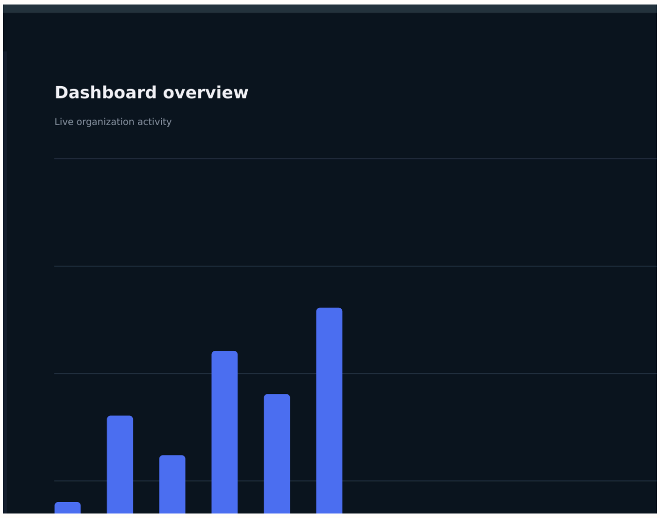
Q01 Qt Operator Architecture · A07 Canonical Specialist Contracts

WebRecon evidence map

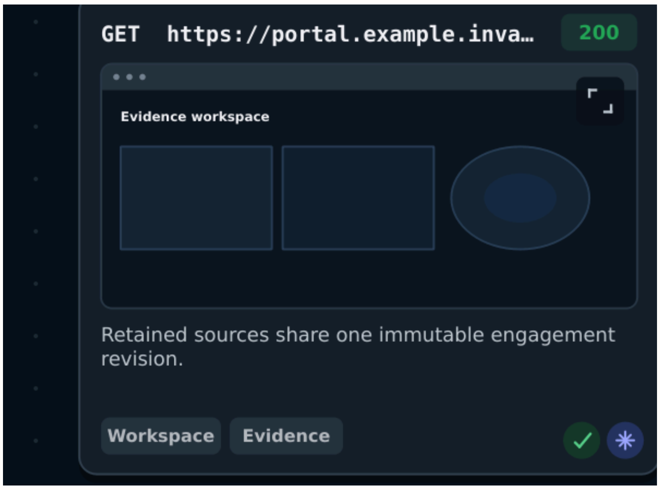
WebRecon accumulates page and response evidence under stable identities. Headers, technologies, screenshots, and review history enrich existing cards. Text updates preserve geometry and camera state; topology changes trigger layout.



Six linked evidence records · Qt capture · synthetic fixture



Retained source preview · synthetic chart



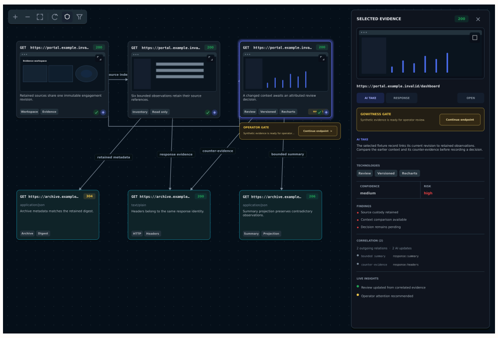
Source card · detail

Web evidence linked by navigation and response relationships.

QT CAPTURES · SYNTHETIC FIXTURES

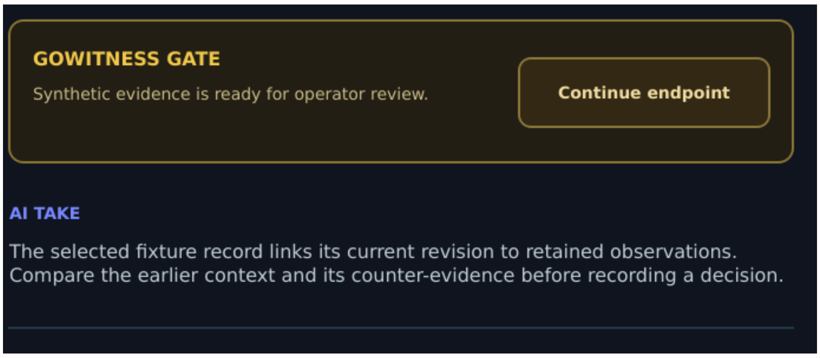
Endpoint review and hypothesis gates

Endpoint review binds an exact gate identity and revision while independent work continues. Repeated decisions are idempotent. The Bounty Hunting Gate (BHG) separately holds a diagnostic hypothesis for explicit policy review. Human-only challenges use their declared response boundary.



OVERVIEW

The selected endpoint exposes its pending browser-review gate.



DETAIL

Qt capture · synthetic fixture

TMG, SCG and BHG ask different questions

Hypothesis matching, prerequisite diagnostics and policy review retain distinct outputs.

TMG
Trigger Methodology Graph
Which hypothesis fits?
Match or partial-match receipt
Diagnostic authority

SCG
Shadow Capability Graph
Which prerequisite is missing?
Hypothetical plan + blockers
Diagnostic authority

BHG
Bounty Hunting Gate
Does policy approve?
Explicit review disposition
Retained gate authority

Canonical evidence and policy context
Each interface consumes declared inputs; no automatic promotion to execution is implied.

Columns: separate contracts · policy review does not prove an external outcome

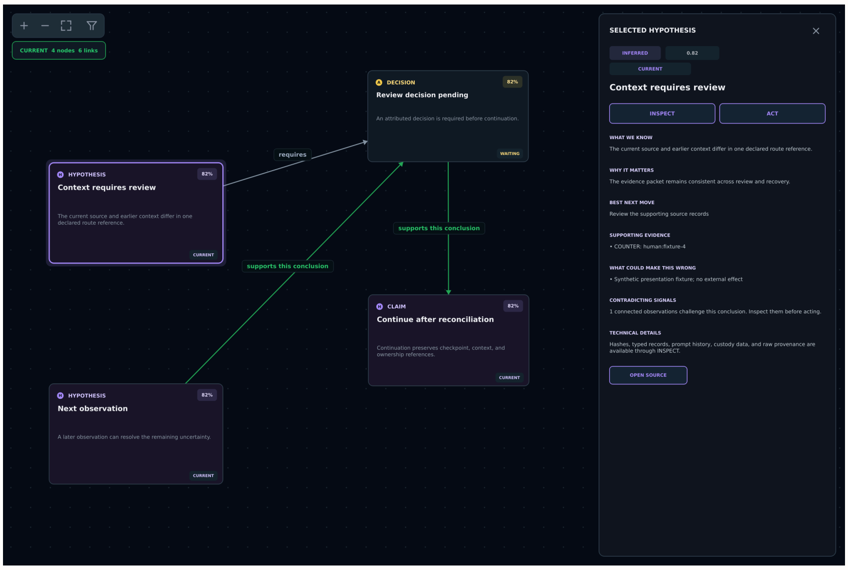
Selected evidence and a pending endpoint review.

QT CAPTURE + CONCEPTUAL DIAGRAM

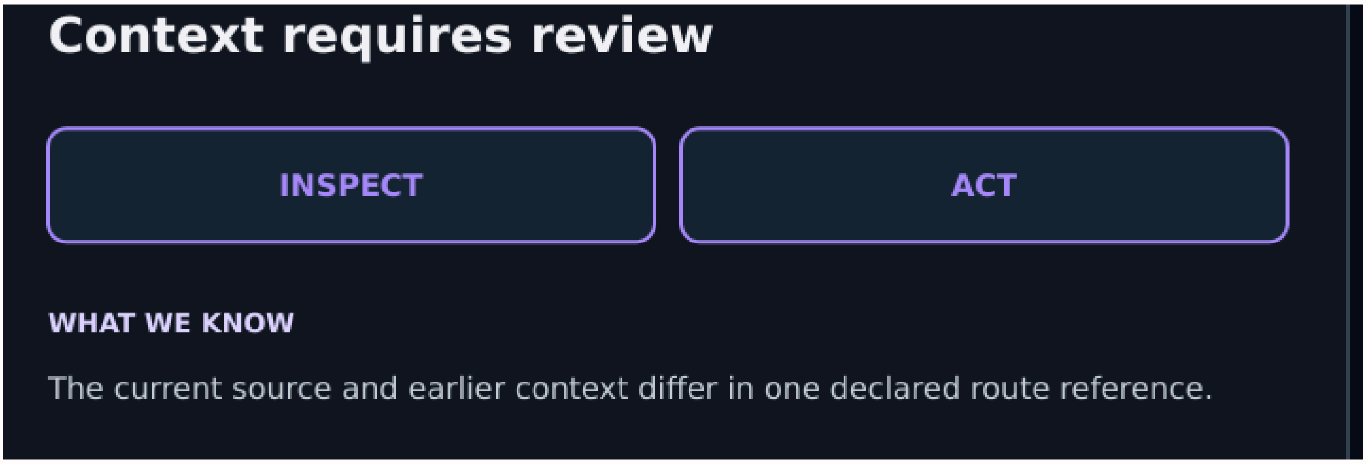
Q03 WebRecon Native Parity · Q05 CAPTCHA Human Gate Contract · M01 Trigger Methodology Graph · G05 RedShades Canonical Vocabulary

Investigation Graph

Investigation Graph presents conclusions with supporting observations, uncertainty, and counter-evidence. Routine records collapse into bounded groups. Canonical detail remains accessible through Inspect, while deterministic attention preserves stable evidence neighborhoods.



Investigation Graph · Qt capture · synthetic fixture



Selected conclusion and its supporting observation · same capture

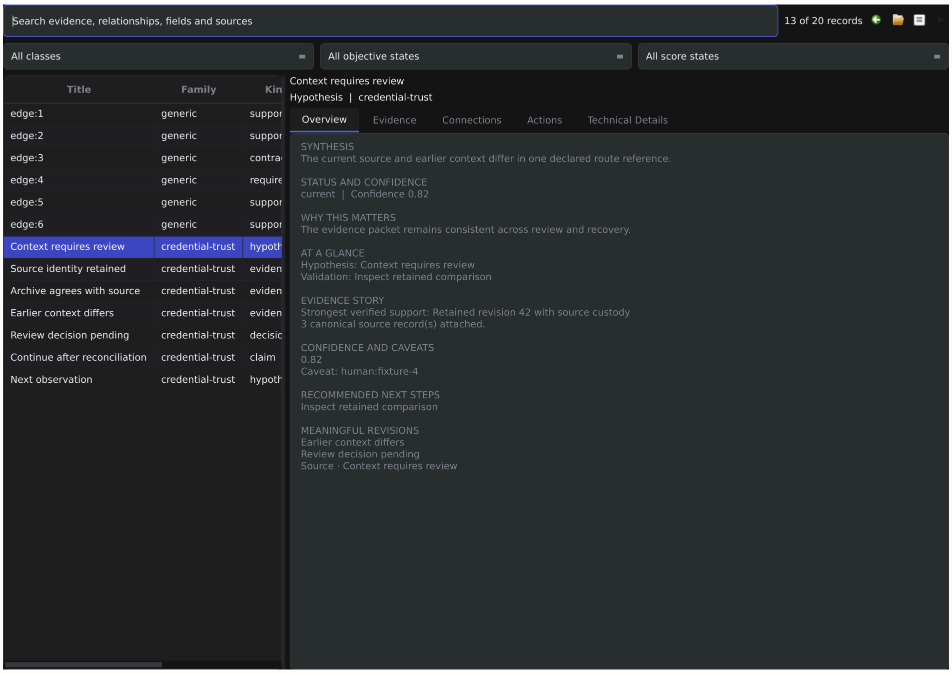
A conclusion remains attached to its supporting records.

QT CAPTURE · SYNTHETIC FIXTURE

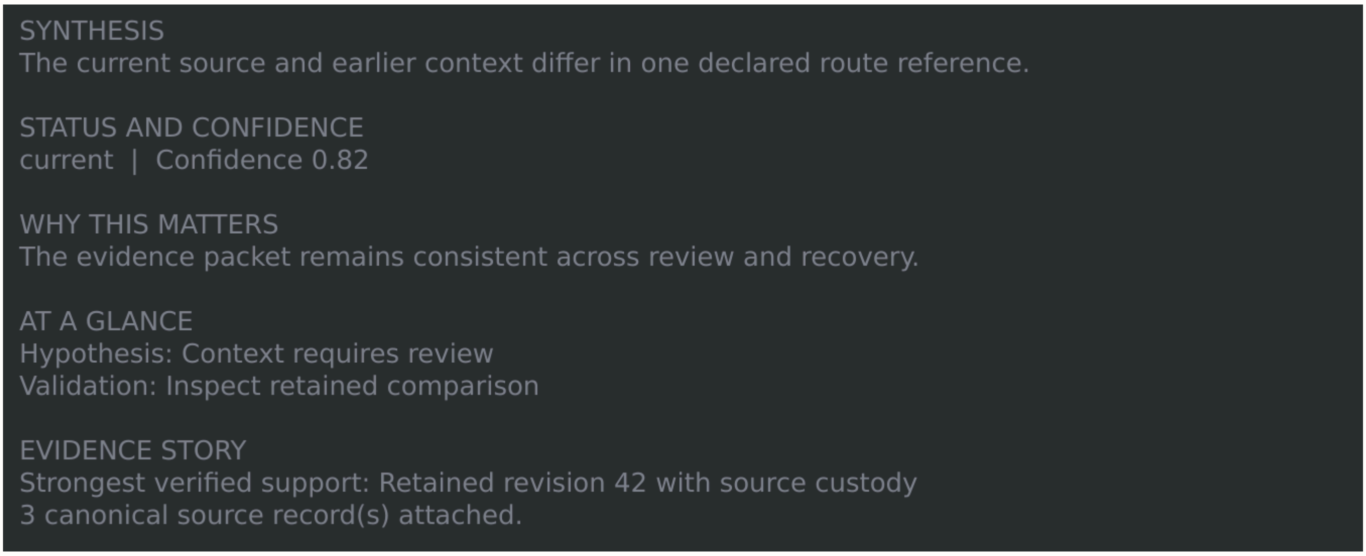
Q04 Investigation Graph and Workbench · A10 Investigation Attention

Investigation Workbench

The Investigation Workbench combines searchable evidence, specialist records, context, and recovery state. Selecting a record loads its owning session once and preserves each child's stable identity. Source navigation retains packet, binary, artifact, and environment coordinates.



Investigation Workbench · Qt capture · synthetic fixture



Canonical detail and provenance · same capture

Canonical record detail preserves source fidelity.

QT CAPTURE · SYNTHETIC FIXTURE

RCC

Recovery and Continuation Center

The Recovery and Continuation Center (RCC) distinguishes interrupted, imported, conflicted, and resumable work. Each action identifies its checkpoint, context, ownership revision, and gate. Pending reconciliation requires attributed acceptance or rejection before lifecycle state changes.

01

Inspect the checkpoint

Retained work, context and exact identity

02

Check ownership

Current generation and resource availability

03

Reconcile

Attributed acceptance or rejection

04

Continue

Revisioned action with a durable receipt

An accepted intent retains its identity across interruption.

CONCEPTUAL DIAGRAM

INPUT	OUTPUT	AUTHORITY BOUNDARY
Checkpoint, context, ownership and gate	Attributed acceptance, rejection or continuation	Reconciliation precedes lifecycle change

IMPLEMENTED CONTRACT

Q04 Investigation Graph and Workbench · A07 Canonical Specialist Contracts

Evaluation levels

Validation progresses from contract review through fixtures, production-orchestration simulation, representative live labs, and broader environmental acceptance. Each level introduces dependencies left controlled by earlier levels. Counterfactual fixtures vary identity, order, absent evidence, and competing candidates.

01

Contract review

Declared behavior and proposed methodology

02

Parser or stage fixture

Supplied inputs and local transitions

03

Orchestration simulation

Production state and coordination under controlled failure

04

Representative live lab

Real tools, processes, services, and operating systems

05

Broader acceptance

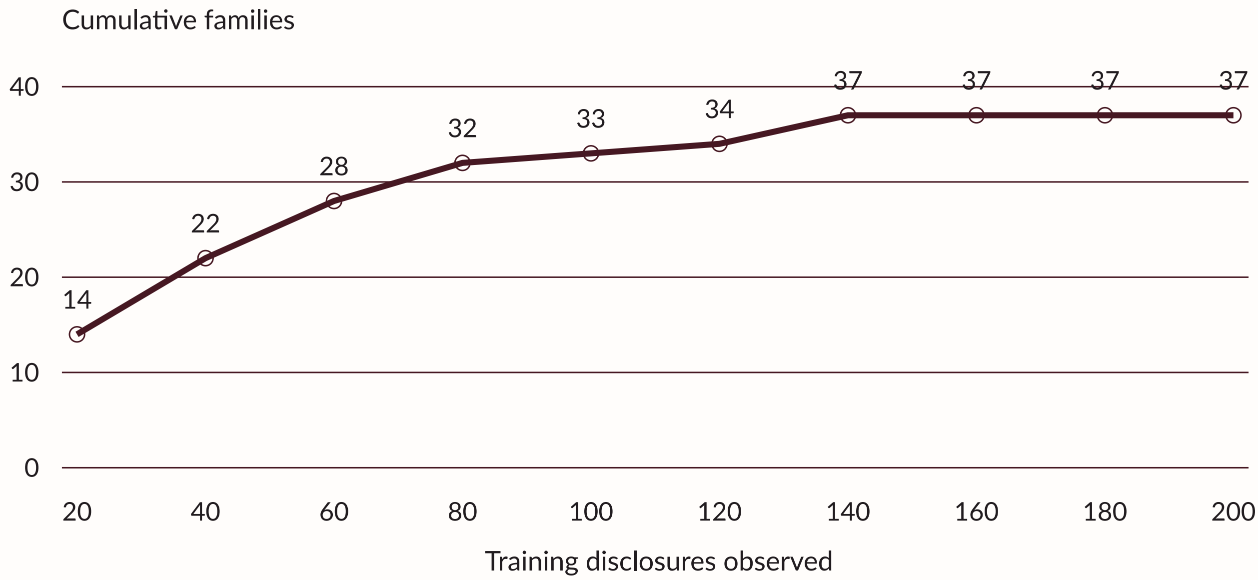
Unusual configurations and sustained integration

Each result is bound to its revision and conditions.

CONCEPTUAL DIAGRAM

Disclosure generalization study

The disclosure study used 200 identity-disjoint training records and 40 held-out records. Training represented 37 families. Three synthetic rotations per candidate yielded 600/600 clean activations. Held-out family coverage was 39/40; live-exercised and admitted candidate counts remained zero.



The curve measures cumulative represented families.

VERSIONED STUDY · 200 TRAINING RECORDS

Advisory abstraction study

Advisory evaluation separates representability, executability, and derivability. The initial checkpoint represented 50/50 discovery records and 24/24 held-out records without analyzer errors or missing primitive classes. Insufficient source evidence left executability and derivability false; 26 held-out records remain untouched.

R

Representability

Can the vocabulary express the procedure?

E

Executability

Are inputs, executor, policy and validation available?

D

Derivability

Does the supplied evidence contain sufficient mechanics?

50/50

discovery records represented

24/24

held-out records represented

A complete vocabulary can coexist with an unresolved procedure.

VERSIONED ADVISORY STUDY

Retained engineering verification

Historical verification covers transactional state, lifecycle reconciliation, specialist interchange, native rendering, and artifact integrity. A simulator report records 18/18 production-orchestration runs and 40/40 reusable-mechanics runs. A later inventory contains eleven authored mechanics and three PostgreSQL production cases.

RETAINED RECORD	SUPPORTED SCOPE
Backend contracts	Transactional state, lifecycle and specialist interchange
Native operator	Rendering, density fixtures and paint telemetry
18/18 runs	Three production discovery scenarios
40/40 runs	Reusable synthetic mechanics
Artifact integrity	Staging, verification and rollback contracts

Only the discovery cases traverse their matching production handlers.
HISTORICAL VERIFICATION

E03 691 Verification Record · V03 Simulator Execution Report

What the architecture has already demonstrated

Reported autonomous completions show that the chain can sustain long sequences beyond one model horizon. Retained simulations and engineering contracts support individual mechanisms; exact field logs remain a separate evidence class and are not reconstructed from narrative.

REPORTED AUTONOMOUS OUTCOME	EVIDENCE CLASS
P.O.O	Reported completion without manual intervention
Hard Windows CTF	Reported completion without manual intervention
Insane Windows HTB lab	Reported completion without manual intervention
Approximately two easy Linux/web labs	Reported completions without manual intervention

The reported outcomes remain separate from CI and simulation receipts.

REPORTED FIELD OUTCOMES

What remains unproven

General reliability across unseen environments, the complete planned QLoRA bank, consistent weak-model parity with stronger agents, broad ACP convergence and full AEP hand-back still require current clean receipts and representative evaluations. The architecture is designed for those goals; the catalog does not present them as finished facts.

BOUNDARY	REMAINING LIMITATION
Legacy adoption	Canonical scheduler and AI integration remains incomplete
Authority	Some legacy stages follow a single launch approval
Adapter breadth	Authentic exports and representative environments remain necessary
Recovery	Projection outages, browser descendants, and unusual environments
AEP / research	Unfamiliar-failure convergence and wider generalization remain unproven

Implementation and environmental evidence have different coverage.

CONCEPTUAL DIAGRAM

A01 Authoritative Engagement Architecture · E05 Operator Frontend Evidence · G07 Historical Chain Assessment · M01 Trigger Methodology Graph · M03 Shadow Capability Graph · R04 Autonomic Engineering Plane Contract

Development direction: measurable intelligence leverage

The decisive metric is not whether AI appears in the loop. It is whether a cheaper bounded model, supported by deterministic state and reusable specialists, reaches the same validated frontier with fewer repairs, no lost feedback and recoverable ownership.

01

Canonical adoption

Scheduler, supervised AI, and repaired CI contracts

02

Environmental verification

Ownership loss, outages, archived state, and adapter fidelity

03

Repair-loop evaluation

Incident history, independent validation, and hand-back

04

Research generalization

Unrelated evidence and unchanged held-out cohorts

RedShades retains explicit acceptance criteria as coverage expands.

CONCEPTUAL DIAGRAM

E04 CI Evidence and Cache Custody · R04 Autonomic Engineering Plane Contract · M02 Disclosure Generalization Study · M04 Advisory Abstraction Research

Acronym and subsystem directory

FFI: Foreign Function Interface. ABI: Application Binary Interface. MCP: Model Context Protocol.

IDENTIFIER	FULL NAME	NARRATIVE
AEP	Autonomic Engineering Plane	22
ACP	Adaptive Capability Pipeline	23
PLO	Phase-Level Optimization	23
TMG	Trigger Methodology Graph	29
SCG	Shadow Capability Graph	29
BHG	Bounty Hunting Gate	35
CES	Canonical Engagement State	06
IGS	Immutable Generation Store	07
EVC	Evidence Verification and Custody	10
ECT	Execution Context Timeline	11
SIC	Specialist Interchange Contracts	12
DTE	Deterministic Transformation Engine	14
WFS	Work Frontier Scheduler	16
PLS	Progress Lease Supervision	16
ADS	AI Decision Service	17
TAC	Transactional Action Coordinator	09
RCK	Runtime Composition Kernel	19
ISW	Isolated Stage Workers	18
RCC	Recovery and Continuation Center	38
SPR	Specialist Plugin Registry	29

Authority and interface index

Schemas retain existing source identities, including EngagementState, StageExecutionEnvelope, StageResultEnvelope, ExecutionContextFrame, AIDecisionReceipt and ReconciliationResult.

OWNER	INPUT / INTERFACE	RETAINED OUTPUT	NARRATIVE
CES / IGS	Expected revision and complete state	Immutable generation, hashes, head	06–07
TAC	Prepared intent and typed executor	Execution/action receipts, observed outcome	09
EVC / SIC / ECT	Artifact, result and factual frame	Custody, coordinates, context lineage	10–13
WFS / PLS / ADS	Evidence, health and decision packet	Eligibility, leases, validated receipt	16–17
ISW / RCK / switchboard	Artifact, input hash and generation	Fenced result, provider lease, custody	18–21
AEP / ACP	Repair incident / capability contract	Distinct admitted candidate histories	22–23
Go/Qt FFI	Versioned snapshots and commands	Bounded view and attributed intent	28
Review gates / RCC	Exact record, owner and gate revision	Decision, reconciliation, continuation	35, 38

A01 Authoritative Engagement Architecture · A07 Canonical Specialist Contracts · R01 Runtime Composition Kernel · R02 Runtime Lifecycle Adoption · G03 Native FFI/API Surfaces · Q04 Investigation Graph and Workbench

Receipt and verification index

EVIDENCE RECORD	ESTABLISHES	LIMIT / STATUS	NARRATIVE
Generation hashes	Complete content and lineage	Separate from process ownership	07
Action intent and receipt	Accepted dispatch identity	Outcome needs observation	09
Raw digest and coordinates	Artifact custody	Separate from semantic interpretation	10, 13
AI decision receipt	Revision, validator and typed result	No scheduler mutation authority	17
Worker lease / generation	Current result ownership	Legacy handlers retain narrower replacement	18
Historical operator CI	Rendering and fixture measurements	July 2026 build-specific evidence	27, 42
Go run 1599	Earlier gates passed; three root failures	Repairs await a new pushed result	25
Simulator report	18/18 production and 40/40 mechanics runs	Bounded synthetic inventories	42
Reported autonomous outcomes	Historical field statements	Exact retained logs unavailable	43

A01 Authoritative Engagement Architecture · A07 Canonical Specialist Contracts · E03 691 Verification Record · E04 CI Evidence and Cache Custody · E05 Operator Frontend Evidence · V03 Simulator Execution Report

Evaluation and maturity index

AREA	EVIDENCE OR CURRENT STATUS	REMAINING ACCEPTANCE	NARRATIVE
Canonical architecture	Implemented contracts	Complete legacy adoption	06–18, 44
Specialist families	Typed interchange and fixtures	Authentic exports and representative environments	12–15, 29
Native operator	Historical captures and measurements	Current visual CI and broader display hosts	27, 30–38
TMG study	37 families; 39/40 held-out coverage	Live exercise and admission remain zero	40
Advisory study	50/50 discovery; 24/24 held-out represented	Derivability/executability; 26 untouched records	41
AEP	Implementation stabilizing	Complete repair and verified hand-back	22, 44–45
Legacy approval	Broad launch approval in some paths	Individual high-impact stage gates	35, 44
Environment recovery	Bounded retained evidence	Archives, projections, browsers, unusual hosts	44–45

E04 CI Evidence and Cache Custody · E05 Operator Frontend Evidence · G07 Historical Chain Assessment · M02 Disclosure Generalization Study · M04 Advisory Abstraction Research · R04 Autonomic Engineering Plane Contract

References and technical review

A private technical review can follow a synthetic RedShades engagement through evidence inspection, a review decision, interruption, and recovery alongside its retained contract and verification record.